Version History:

Date	Version	Modifier	Change Reason	Description
2015.10.02	b0.1	Gaujei		
2015.10.05	b0.2	Golem	Add functions based on current hardware.	Add Fan, Charge LED command. Motion command based on motor. Add IR, AGM, LED, Battery, Charge IC, Fan feedback.
2015.10.06	b0.21	Golem	Functions used for factory or calibration.	Clear Alarm for reset the status. Motion command for task command. Set offset value for sensors. Set emerge stop value for sensors. Write setting values into flash.
2015.10.08	b0.3	杰哥		
2015.10.12	b0.31	Golem	Unified the format. New function for motor life.	Add new request in Request Extra for motor life info. and feedback as well.
2015.10.13	b0.32	Golem	Check motor functions.	Rearrange controller number structure for easier use. Modify motor controller mode, motor status feedback.
2015.10.27	b0.40	GauJei	Command package	Command package implement.
2015.10.30	b0.41	GauJei	Feedback package	Feedback package implement.
2015.10.31	b0.42	GauJei	Add description for initial	Modify the method of terminal communities and add description at Request Extra info.
2015.12.17	b0.43	GauJei	Add a ping function	Modify the length of

				Request Extra information
2016.01.07	b0.44	GauJei	Modify the sonar date size	Request from BSP.
2016.01.18	b0.45	Ting-Ying/ Ben	a. Modify time-stamp size from 2 to 4 bytes. b. Add a new command for velocity control	a. Request from vision team b. Set velocity for both wheels
2016.02.01	b0.46	GauJei	Check peripheral status	Add the status for Drop IR and IMU
2016.02.15	b0.47	Ben	Adjust the command structure for basic sensor data	Redefine the command for encoder, odometry and avoidance target.
2016.03.02	b0.48	ZiYao/Ben	Ben: Modify data length of wheel encoder ZiYao: a. Update Neck Status Add a new command for neck velocity ctrl.b. Add Docking and User Tracking command	Ben: Modify data length from 4 to 8 bytes ZiYao:
2016.03.14	b0.49	Ben	a. Hardware version b.modify the size of data for sonar and drop	
2016.04.14	b0.50	ZiYao	a. add a new command for base position control b. add three status	a. Request from ADC team
2016.04.19	b0.51	ZiYao	Add new commands for neck smooth path control	
2016.04.21	b0.52	Golem	a. add modify Controller Limit	for parameter write / read
2016.04.21	b0.53	Golem/Ben	a. add Calibration Write(Command) b. add Calibration Return(Feedback)	for factory test
2016.04.27	b0.54	Golem/ Danny/Ben	a.modify motor control info b. add docking status c. add new command for power on docking station	for motion control parameter set/get from android
2016.04.29	b0.55	ZiYao	a. modify base task control	

			b. add motion vendor command	
2016.05.11	b0.56	ZiYao	a. Modify mobile status b. Modify unit in neck velocity control and Controller output limit c. Change motor number in Controller output limit & Controller Offset & Controller Gain	
2016.05.12	0.57	Ben	a. Shift the transfer length when transfer of data is equal to the max packet size.	For USB transfer
2016.05.16	0.58	Danny	a. Modify Docking IR feedback data form(0x04) b. Modify command 0x20 bit 2&3 for sonar switch	
2016.05.18	0.59	Golem	Fix typo	
2016.05.26	0.60	Danny	a. Modify basic sensor data(sonar)	
2016.05.27	0.61	Danny	a. Add Error Code	
2016.06.14	0.62	ZiYao	a. Modify docking status	
2016.06.20	0.63	Golem	a. Add Drop Sensor rate return. b. Add Drop IR crosstalk, rate limit in flash.	
2016.06.29	0.64	Golem	a. Add calibration close return command(case 0xFF)	For ER factory SW lost log
2016.7.13	0.65	ZiYao	a. Modify Base Velocity Control ID0x01	
2016.7.14	0.66	ZiYao	a. Update availability of IMU data b. Update unit of IMU data	
2016.7.21	0.67	Danny	a. Change Basic sensor data DockingStatus bit 7 to ContactDockingStation	
2016.8.9	0.68	ZiYao	a. Add feedback packets of motor current	

2016.8.26	0.69	Danny	a. Neck status bit 6 & 7 change to Absolute Encoder Yaw Status & Absolute Encoder Pitch Status b. Change over current and over temputure bit arrangement.	
2016.8.30	0.70	Danny	a. Remove Neck status bit 2 & 3 b. Change Neck status bit 4,5,6,7 Absolute Encoder Yaw FW Error Absolute Encoder Pitch FW Error Absolute Encoder Yaw HW Error Absolute Encoder Pitch HW Error	
2016.09.01	0.71	Golem	a. OCP,OLP flag b. VelBusy flag c. clear unused Calibration Command	
2016.9.05	0.72	ZiYao	Add reading/writing command of drop 4 line segment	
2016.9.06	0.73	ZiYao	a. add a new commend which request control information b. add 3 feedback package (neck/wheel trajectory, PWM)	
2016.09.14	0.74	Golem	a. add position control performance.	
2016.09.20	0.75	Golem	a. add Neck Return for saving time.	
2016.09.29	0.76	ZiYao	a. Add request flag of motor current (bit 8) in “Request Feedback Info”(ID 0x08)	Fix v0.68
2016.10.12	0.77	Golem	a. Add Battery Info.	
2016.11.9	0.79	Danny	a. Add drop IR type	
2016.11.18	0.80	Golem	a. Add drop IR type for dev	
2016.11.23	0.81		a. Add motor controller info. for	

			BSP	
2016.12.15	0.82	Lucas	a. Modified dropIR flag	B & D region
2016.12.21	0.83	Danny	a. Add R/W for drop IR calibration raw data	
2017.01.05	0.84	HC Chou	a. Increase the number of input parameters of motion_SetTurningCoefficient function b. A rule-based fuzzy system is designed for avoidance algorithm instead of fuzzy-apf fusion approach	
2017.01.05	0.85	ZiYao	a. Remove smooth path function b. Fix v0.84	1. Command ID 0x0E & 0x0F 2. Smooth path buffer flag in Basic Sensor Data
2017.03.02	0.86	ZiYao	a. Add avoidance type in Base Velocity Control & Base Position Control b. Update user tracker c. Add drop flag (motion level) instead of battery status in the basic sensor data	
2017.05.08	0.87	Golem	Modify gyro update format	
2017.05.23	0.88	Golem	Add Low-power Switch Command	Merge from Lowpower branch
2017.05.24	0.89	Danny	Modify type of low-power command and return	
2017.07.25	0.90	Golem	Update Calibration Command/Feedback Type for multi-segment	
2017.10.11	0.91	Golem	Add Write Flash command, and request calibration checksum command/feedback.	For recovering calibration data.

Overview

- Structure of Byte stream

- Command packets
- Feedback packets
- Function Classification

Structure of Byte stream

A byte stream can be divided into 4 fields; Headers, Length, Payload and Checksum.

Name	**Header 0**	**Header 1**	**Length**	**Payload**	**Checksum**
Size(Byte)	1	1	1	N	1
Description	0xAA (Fixed)	0x55 (Fixed)	Size of payload in bytes	Described below	XOR'ed value of every bytes of byte stream except headers

Header: AA 55

Length: Default size of this field is 1 byte.

Command payload is without sub-payloads.

Feedback Payload is included with sub-payloads.

Payload				
Sub-Payload 0	Sub-Payload 1	Sub-Payload 2	…	Sub-Payload N-1

Sub-payload: it can be divided into three parts; Header, Length and Data.

Name	Header	Length	Data
Size(Byte**)**	1	1	N
Description	Predefined identifier	Size of data in byte(s)	Described below

Checksum:

’OR'ed value of every bytes of byte stream that except headers. Checksum process must ensure the integrity of byte streams.

Serialization-Deserialization:

The process of converting a data structure that into a byte stream. Deserialization is reversal process. Each data types are serialized and desterilized by `LSB-First Order`. It means Least Significant Bytes will come first on the byte stream.

Command packets

Command Identifier

ID	Name	Description
0x01	Base Velocity Control	Set linear velocity and rotation speed for mobile base.
0x02	Wheel Velocity Control	Set velocity for both wheels
0x03	Neck Velocity Control	Set neck velocity command for neck joint
0x04	Neck Position Control	Set angle command for neck joint
0x05	Motor PWM Output	Set PWM value out to a motor drive
0x06	Base Task Size	Set task command size
0x07	Base Task Control	Set task command data (x, y, theta)
0x08	Request Feedback Info	Request sensor information
0x09	Request Extra Info	Request version information
0x0A	Request Motor Control Info.	Get motor information
0x0B	Base Position Control	Set wheel motion in base space
0x0C	User Tracking	Set neck and wheel action to track user
0x0D	Docking	Start docking
~~0x0E~~	~~Smooth Path Size~~	~~Set path command size~~
~~0x0F~~	~~Smooth Path Control~~	~~Set path command data~~
0x11	Controller Output Limit	Set Upper and lower limits
0x12	Controller Offset	Input offset of a velocity command
0x13	Controller Gain	Set PID gain of a velocity controller.
0x14	Request Controller Info	Request current controller information.
0x1A	Motion Vendor	Motion vendor access
0x1B	Request Control Info	Request controller information
0x20	Request Peripheral Info	Get peripheral information of robot.
0x21	Peripheral Control	Set low-power control mode

0x23	Control Test	The test by firmware
0x25	Vendor Driver IO	Vendor driver I/O access
0x30	Sensor Offset	Sensor offset for calibration.
0x31	Sensor EMG Value	The level of EMG for Sensor.
0x32	Request Sensor Info	Get Sensor offset/EMG information
0x33	Battery Info.	Get Battery Info. From Android
0x40	Calibration Write	Calibration trigger or write value
0x41	Request Calibration Info	Get calibration values
0x42	Write Calibration data to Flash	Write Calibration data directly to Flash
0x50	Docking Station Charging control	Set Docking Station to charge or not

- **Base Velocity Control**

Velocity Representation

Motion	Speed(mm/s)	Radius(mm)
Pure Translation	Speed:+ forward , - backward	0
Pure Rotation	ω * b/ 2 ; w in rad/sec b: indicates the length between the center of the wheels	1: turn left , -1: Right
Translation + Rotation	Speed * (Radius + b/ 2) / Radius, if Radius > 1 Speed * (Radius - b/ 2) / Radius, if Radius < -1	Radius

	Name	Size	Value	Description
Header	**Id**entifier	1	0x01	Fixed
Length	Size of data field	1	0x08	Fixed
Data	Velocity speed	2	-500 ~ 500	Unit : mm/s
	Rotation speed	2	-2664 ~ 2664	Unit : 0.1 deg/s
	Acceleration time	2	5 ~ 65535	Unit : ms
	Type of avoidance	2		0x01 : ByPass 0x02 : Normal 0x03 : Simple 0x04 : Corridor 0x05 : Door

Note : this command/function contain drop protection

- **Wheel Velocity Control**

Set motor velocity in mm/sec.

	Name	Size	Value	Description
Header	Identifier	1	0x02	Fixed
Length	Size of data field	1	0x08	Fixed
Data	Speed Left	2	-500 ~ 500	Unit : mm/sec
	Speed Right	2	-500 ~ 500	Unit : mm/sec
	Time Left	2		
	Time Right	2		

- **Neck Velocity Control**

Set motor velocity in 0.1°/s

	Name	Size	Value	Description
Header	Identifier	1	0x03	Fixed
Length	Size of data field	1	0x08	Fixed
Data	Velocity Yaw	2		Unit : 0.1°/s,
	Velocity Pitch	2		Unit : 0.1°/s
	Time Yaw	2	0~65535	Unit : ms
	Time Pitch	2	0~65535	Unit : ms

- **Neck Position Control**

Set motor position in degree or length.

	Name	Size	Value	Description
Header	Identifier	1	0x04	Fixed
Length	Size of data field	1	0x08	Fixed
Data	Position Yaw	2	-450~450	Unit : 0.1°
	Position Pitch	2	-150~550	Unit : 0.1°
	Time Yaw	2	5~65535	Unit : ms
	Time Pitch	2	5~65535	Unit : ms

- **Motor Drive PWM Output**

Set PWM value out to a motor drive.

	Name	Size	Value	Description
Header	Identifier	1	0x05	Fixed

Length	Size of data field	1	0x03	Fixed
Data	Controller select	1		0:disable, 1:enable Bit 0: right wheel Bit 1: left wheel Bit 4: neck pitch Bit 5: neck yaw
	PWM output	2	-3600~3600	1 Unit : 0.028 of full duty cycle

- **Base Task Size**

Data size of the following base controller in task space.

	Name	Size	Value	Description
Header	Identifier	1	0x06	Fixed
Length	Size of data field	1	0x01	Fixed
Data	Data size	2		

- **Base Task Control**

Set task command data (x, y, theta)

	Name	Size	Value	Description
Header	Identifier	1	0x06	Fixed
Length	Size of data field	1	0x0A	Fixed
Data	Index of the data	2		
	Task space data x	2		Unit : mm
	Task space data y	2		Unit : mm
	Task space data theta	2		Unit : 0.1°
	Type of avoidance	2		0x01 : ByPass 0x02 : Normal 0x03 : Simple 0x04 : Corridor 0x05 : Door

- **Request Feedback Info**

	Name	Size	Value	Description
Header	Identifier	1	0x08	Fixed

Length	Size of data field	1	0x02	Fixed
Data	Request flags	2		Set the flags to request feedback sensor data. Bit 0: Drop IR Bit 1: Sonar Bit 2: Docking Bit 3: G sensor Bit 4: Gyro Bit 5: E-compass Bit 6: Motor temperature Bit 7: Drop IR rate Bit 8: Motor Current Bit 9: Drop IR type

- **Request Extra information**

Get version information.

	Name	Size	Value	Description
Header	Identifier	1	0x09	Fixed
Length	Size of data field	1	0x01	Fixed
Data	Request switch	1		Set the flags to request extra data. Bit0: Hardware Version Bit1: Firmware Version Bit2:Drop IR type Bit3:Unique Device ID Bit4:Calibration data checksum

Note that it will trigger the feedback package start to update for initial when set the request switch to 1, and set to 0 to stop the data/status update.

- **Request Motor Control information**

	Name	Size	Value	Description
Header	Identifier	1	0x0A	Fixed
Length	Size of data field	1	0x01	Fixed
Data	Request flags	1		1 : motor life info 2 : motor servo info 3 : motor ctrl type

- **Base Position Control**

	Name	Size	Value	Description

Header	Identifier	1	0x0B	Fixed
Length	Size of data field	1	0x0A	Fixed
Data	Position X	2		Unit : mm
	Position y	2		Unit : mm
	Position Theta	2		Unit : 0.1°
	time	2		Unit : ms
	Type of avoidance	2		0x01 : ByPass 0x02 : Normal 0x03 : Simple 0x04 : Corridor 0x05 : Door

- **User Tracking**

	Name	Size	Value	Description
Header	Identifier	1	0x0C	Fixed
Length	Size of data field	1	0x08	Fixed
Data	Velocity Yaw	2		Unit : 0.1°/s
	Velocity Pitch	2		Unit : 0.1°/s
	Body linear velocity	2		Unit : mm/s
	Body angular velocity	2		Unit : 0.1deg/s

- **Docking**

	Name	Size	Value	Description
Header	Identifier	1	0x0D	Fixed
Length	Size of data field	1	0x04	Fixed
Data	Mode Select	2	0x00~0x02	0x00 : stop process 0x01 : start docking process 0x02 : start undocking process
	Initial Direction	2		

- **Smooth Path Size**

	Name	Size	Value	Description
Header	Identifier	1	0x0E	Fixed
Length	Size of data field	1	0x01	Fixed

Data	Data size	2	0x0000~0xFFFF	Data size of the following neck smooth path control

➢ **Smooth Path Control**

	Name	Size	Value	Description
Header	Identifier	1	0x0F	Fixed
Length	Size of data field	1	0x0A	Fixed
Data	Position Yaw	2		Unit : 0.1°
	Position Pitch	2		Unit : 0.1°
	Time Yaw	2		Unit : ms
	Time Pitch	2		Unit : ms
	Data Index	2		The number of data command

➢ **Turning Coefficient**

	Name	Size	Value	Description
Header	Identifier	1	0x10	Fixed
Length	Size of data field	1	0x0A	Fixed
Data	Distance of front obstacles	2		Unit : mm
	Distance of front left obstacles	2		Unit : mm
	Distance of front right obstacles	2		Unit : mm
	Distance of left obstacles	2		Unit : mm
	Distance of right obstacles	2		Unit : mm

➢ **Controller Limit**

	Name	Size	Value	Description

Header	Identifier	1	0x11	Fixed
Length	Size of data field	1	0x06	Fixed
Data	type	1		0x00:Velocity Control 0x01:Position Control 0x02:Acceleration Value
	Controller number	1		0 : neck yaw 1 : neck pitch 2 : left wheel 3 : right wheel
	Upper limit value	2	int16	Max velocity(mm/s) (0.1°/s)for Velocity Control Limit position(0. 1°) for Position Control Max Acceleration(mm/s^2) (0.1° /s^2) for Acceleration Value
	Lower limit value	2	int16	Min velocity(mm/s) (0.1°/s) for Velocity Control Limit position(0.1°) for Position Control Min Acceleration(mm/s^2) (0.1° /s^2) for Acceleration Value

Controller Offset

	Name	Size	Value	Description
Header	Identifier	1	0x12	Fixed
Length	Size of data field	1	0x03	Fixed
Data	type	1		0x00:Velocity Control 0x01:Position Control
	Controller number	1		0 : neck yaw 1 : neck pitch 2 : left wheel 3 : right wheel
	offset value	2		

Controller Gain

Set PID gain of controller of robot.

	Nam	Size	Value	Description

Header	Identifier	1	0x13	Fixed
Length	Size of data field	1	0x08	Fixed
Data	type	1		0x00:Velocity Control 0x01:Position Control
	Controller number	1		0 : neck yaw 1 : neck pitch 2 : left wheel 3: right wheel
	P gain	2		uint16_t; 1 Unit = 0.01
	I gain	2		uint16_t; 1 Unit = 0.1
	D gain	2		uint16_t; 1 Unit = 0.0001

- ## Controller Info

Request Controller information of robot.

	Name	Size	Value	Description
Header	Identifier	1	0x14	Fixed
Length	Size of data field	1	0x03	Fixed
Data	Controller type	1		0x00:Velocity Control 0x01:Position Control 0x02:Acceleration Value
	Controller number	1		0 : neck yaw 1 : neck pitch 2 : left wheel 3 : right wheel
	Information type	1		0x01: output limit 0x02:Offset 0x03:PID Gain

- ## Motion Vendor

Set motion test function

	Name	Size	Value	Description
Header	Identifier	1	0x1A	Fixed
Length	Size of data field	1	0x04	Fixed
Data	address	2		Byte 0 : type Byte 1 : Address

	data	12		

➢ Request Control Information

Get control information.

	Name	Size	Value	Description
Header	Identifier	1	0x1B	Fixed
Length	Size of data field	1	0x02	Fixed
Data	Request switch	2		Set the flags to request command. Bit0: neck trajectory Bit1: wheel trajectory Bit2: motor PWM

➢ Request Peripheral Information

Get peripheral information.

	Name	Size	Value	Description
Header	Identifier	1	0x20	Fixed
Length	Size of data field	1	0x01	Fixed
Data	Request switch	1		Set the flags to request command. Bit0: Drop IR info Bit1:IMU info Bit2:Sonar info ON Bit3:Sonar info OFF

➢ Peripheral Control

Set low-power control mode

	Name	Size	Value	Description
Header	Identifier	1	0x21	Fixed
Length	Size of data field	1	0x02	Fixed
Data	Request Type	1		Set target items. Bit0:sonar process disable

				Bit1:I2C process disable Bit2:IMU sleep mode Bit3:Drop IR disable Bit4:USB update disable Bit5:sonar HW disable Bit6:Request Feedback
	Mode select	1		Set / Reset. Bit0:sonar process disable Bit1:I2C process disable Bit2:IMU sleep mode Bit3:Drop IR disable Bit4:USB update disable Bit5:sonar HW disable

- **Velocity Control Test**

The test item is setting by firmware.

	Name	Size	Value	Description
Header	Identifier	1	0x23	Fixed
Length	Size of data field	1	0x0D	
Data	test item	1		0 : velocity control 1 : velocity profile
	No. of controller	1		
	Launch	1		0 : stop 1 : start
	Target velocity	2	-2000 ~ 2000	Unit : mm/s, default : 0
	Max limit	2	500~2000	Unit : mm/s, default : 500
	Distance	2	0~1000	Unit : cm, default : 300
	Acceleration Time	2	0~3000	Unit : ms, default : 2000
	Test cycles	2	1~65535	Unit : times, default : 1

- **Vendor Driver IO**

Vendor driver control setting.

	Name	Size	Value	Description
Header	Identifier	1	0x25	Fixed
Length	Size of data field	1	0x06	Fixed
Data	address	3		Byte 0-1 : vendor id

				Byte 2 : device id
	data	3		

- **Sensor Offset**

	Name	Size	Value	Description
Header	Identifier	1	0x30	Fixed
Length	Size of data field	1	0x04	Fixed
Data	Sensor type	1		0x00: Drop IR 0x01: IR Range Finder 0x02: Dock IR 0x03: Sonar 0x04: Temperature 0x05: Drop IR crosstalk 0x06: Drop IR rate limit
	sensor serial number	1		0x01: sensor #1 0x02: sensor #2 0x03: sensor #3 0x04: sensor #4:
	Offset Value	2		

- **Sensor EMG value**

	Name	Size	Value	Description
Header	Identifier	1	0x31	Fixed
Length	Size of data field	1	0x04	Fixed
Data	Sensor type	1		0x00: Drop IR 0x01: IR Range Finder 0x02: Dock IR 0x03: Sonar 0x04: Temperature
	sensor serial number	1		0x01: sensor #1 0x02: sensor #2 0x03: sensor #3 0x04: sensor #4 :
	EMG Value	2		

- **Sensor Info**

Request the sensor offset/EMG value.

	Name	Size	Value	Description
Header	Identifier	1	0x32	Fixed
Length	Size of data field	1	0x03	Fixed
Data	Sensor type	1		0x00: Drop IR 0x01: IR Range Finder 0x02: Dock IR 0x03: Sonar 0x04: Temperature
	Sensor serial number	1		0x01: sensor #1 0x02: sensor #2 0x03: sensor #3 0x04: sensor #4 :
	Sensor information type	1		0x01: Sensor offset 0x02: EMG value

- **Battery Info**

Get battery information.

	Name	Size	Value	Description
Header	Identifier	1	0x33	Fixed
Length	Size of data field	1	0x04	Fixed
Data	AC_Flag	1		0x00: battery, 0x01: charger
	Type	1		0x00: Voltage
	Value	2		Voltage: mV

- **Calibration Write**

	Name	Size	Value	Description
Header	Identifier	1	0x40	Fixed
Length	Size of data field	1	0x04	Fixed
Data	calibration type	1		0x01:Gsensor 0x03:E-compass 0x04:Neck 0x06:Drop_offset 0x07:Sonar 0x0A:Drop_crosstalk 0x0C:Drop 4 line segment (distance)

				0x0D:Drop 4 line segment (rate) 0x0E:Position Controller Performance 0x10:Drop IR LEVEL3 distance ~~0x11:Drop IR LEVEL3 rate(reserve)~~ 0xFF:close return
	Calibration number	1		0x00: #1 0x01: #2 0x02: #3 0x03: #4 0x04: #5 0x05: #6
	Calibration data	2	Int16	

➢ Calibration Info

	Name	Size	Value	Description
Header	Identifier	1	0x41	Fixed
Length	Size of data field	1	0x02	Fixed
Data	calibration type	1		0x01:Gsensor 0x02:Gyro 0x05:Wheel 0x06:Drop_offset 0x09:Absolute Encoder 0x0A:Drop_crosstalk 0x0B:Incremental_Encoder 0x0C:Drop 4 line segment (distance) 0x0D:Drop 4 line segment (strength) 0x0F:Neck Return 0x10:Drop IR LEVEL3 distance 0x11:Drop IR LEVEL3 rate 0x13:Drop IR LEVEL3 ID 0xFF:close return
	Calibration number	1		0x00: #1 0x01: #2 0x02: #3 0x03: #4 0x04: #5 0x05: #6

0x0F:Neck Return: number0~3: yaw ZPOS, yaw MPOS, pitch ZPOS, pitch MPOS

➢ Write Calibration data to Flash

	Name	Size	Value	Description
Header	Identifier	1	0x42	Fixed
Length	Size of data field	1	0x04	Fixed
Data	calibration type	1		0x01:Gsensor 0x02:Gyro 0x06:Drop_offset 0x0A:Drop_crosstalk 0x0F:Neck Return 0x10:Drop IR LEVEL3 distance 0x11:Drop IR LEVEL3 rate 0x13:Drop IR LEVEL3 ID
	Calibration number	1		0x00: #1 0x01: #2 0x02: #3 0x03: #4 0x04: #5 0x05: #6
	Calibration data	2	uint16	

- **Docking Station Charging control**

	Name	Size	Value	Description
Header	Identifier	1	0x50	Fixed
Length	Size of data field	1	0x01	Fixed
Data	Charging station Power	1	0x00	0x00:Power off Charging station 0x01:Power on Charging station

Feedback packets

Toby sends below default feedbacks periodically in 50 Hz, when it powered on.

ID	Name	Description	Availability
0x01	Basic Sensor Data	Basic core sensor status/data	**By default**
0x02	Drop IR RAW Data	IR x3	On request

0x03	Sonar Sensor RAW Data	Front x2 , rear x1	On request
0x04	Docking IR RAW Data	Signals from docking station. (ADC x3)	On request
0x05	Accelerometer RAW Data	Accelerometer raw data	On request
0x06	Gyroscope RAW Data	Gyro raw data	On request
0x07	E-compass RAW Data	E-compass raw data	On request
0x08	Drop IR Type	Drop IR Type *5	On request
0x09	Motor Temperature RAW Data	Temperature sensor x4	On request
0x0A	Motor Life	Record moving distance had been done.	On request
0x0B	Peripheral Drop IR	Check drop IR status	On request
0x0C	Peripheral IMU	Check IMU status	On request
0x0D	Drop IR rate	Return rate of drop IR sensor	On request
0x0E	Motor Current	Return current of motor	On request
0x0F	Motor Controller Info.	Return motor controller infomations.	On request
0x10	Hardware Version	Version number of hardware	On request
0x11	Firmware Version	Version number of firmware	On request
0x12	HW drop IR	Drop IR type	On request
0x13	Controller Limit Info	Present offset and output limit of the selected controller	On request
0x14	Controller Offset Info		On request
0x15	Controller PID Info	Servo PID parameters	On request
0x16	Sensor Offset		On request
0x17	Sensor EMG value		On request
0x18	Calibration value	Return calibration status or value	On request
0x19	Peripheral Control Feedback		On request
0x1A	Calibration checksum	Return checksum of calibration data in Flash	On request
0x20	Wheel Encoder	Accumulated encoder data for wheel motor	**By default**
0x21	Neck Encoder	Accumulated encoder data for neck motor	**By default**
0x22	Odometry		**By default**
0x23	Avoid Target	The new target point in x and y axis while obstacle avoidance	**By default**
0x30	Neck trajectory		On request

0x31	Wheel trajectory		On request
0x32	Motor PWM		On request
0xE0	Error Code		**By default**
0xFF	Virtual Data	Use the virtual payload to avoid checksum error	**By default**

- **Basic Sensor Data**

The Basic core sensor data:

	Name	Size	Value	Description
Header	Feedback Identifier	1	0x01	Fixed
Length	Size of data field	1	14	Fixed
Data	Timestamp	4		Timestamp generated internally in milliseconds, It circulates from 0 to 65535/ 2^32
	IR Drop Flag (sensor level)	1		Bit 0 : L45 Drop IR Bit 1 : L70 Drop IR Bit 2 : R45 Drop IR Bit 3 : R70 Drop IR Bit 4 : B60 Drop IR Bit 5 : All Drop IR Bit 6 : B-zone Drop IR Bit 7 : D-zone Drop IR
	Sonar Flag	1		Bit 0 :Right Sonar Bit 1 : Left Sonar Bit 2 : Rear Sonar Bit 3 : Front Right Sonar Bit 4 : Front Left Sonar Bit 5 : Front Center Sonar
	Current Protection	1		Bit 0~3:OCP, over current protection Bit 4~7:OLP, over load protection Bit 0/4: neck yaw Bit 1/5: neck pitch Bit 2/6: left wheel Bit 3/7: right wheel
	Over temperature	1		Bit 0: neck yaw Bit 1: neck pitch Bit 2: left wheel Bit 3: right wheel

				Bit 7 : Battery
	Neck Status	1		Bit 0 : Busy Yaw Bit 1 : Busy Pitch Bit 2 : bit 3 : bit 4 : Absolute Encoder Yaw FW Error bit 5 : Absolute Encoder Pitch FW Error bit 6 :Absolute Encoder Yaw HW Error bit 7: Absolute Encoder Pitch HW Error
	Mobile Status	1		Bit 0 : VelBusy Bit 1 : Error Bit 2 : busy bit 3 : Error Set Odometry bit 4 : Avoidance bit 5 :index0 bit 6 : index1 bit 7: index2
	IR Drop Flag (motion level)	1		Bit 0 : L45 Drop IR Bit 1 : L70 Drop IR Bit 2 : R45 Drop IR Bit 3 : R70 Drop IR Bit 4 : B60 Drop IR Bit 5 : All Drop IR Bit 6 : B-zone Drop IR Bit 4 : D-zone Drop IR
	Docking Status	1		Bit 0 : State 0 Bit 1 : State 1 Bit 2 : State 2 bit 3 : State 3 bit 4 : Busy bit 5 : Success bit 6 : Stop bit 7 : ContactDockingStation
	~~Smooth Path~~ ~~Buffer Flag~~ Route buffer condition	1		Bit 0 : Index1 Bit 1 : Index2 Bit 2 : Index3 bit 3 : bit 4 : bit 5 : bit 6 : bit 7:
	Base Position Control flag	1		Bit 0 : Index1 Bit 1 : Index2 Bit 2 :

				bit 3 : bit 4 : bit 5 : bit 6 : bit 7:

- **Drop IR Information**

	Name	Size	Value	Description
Header	Identifier	1	0x02	Fixed
Length	Size of data field	1	0x0A	Fixed
Data	Drop IR fifth(back 60)	2	0~65535	Unit: mm
	Drop IR fourth(right 70)	2	0~65535	Unit: mm
	Drop IR third(right 45)	2	0~65535	Unit: mm
	Drop IR second(left 70)	2	0~65535	Unit: mm
	Drop IR first (left 45)	2	0~65535	Unit: mm

- **Sonar Sensor Information**

	Name	Size	Value	Description
Header	Identifier	1	0x03	Fixed
Length	Size of data field	1	0x0C	Fixed
Data	Sonar right	2	0~65535	Unit: mm
	Sonar left	2	0~65535	Unit: mm
	Sonar back	2	0~65535	Unit: mm
	Sonar front right	2	0~65535	Unit: mm
	Sonar front left	2	0~65535	Unit: mm
	Sonar front center	2	0~65535	Unit: mm

- **Docking IR RAW Data**

	Name	Size	Value	Description
Header	Identifier	1	0x04	Fixed
Length	Size of data field	1	0x06	Fixed

Data	Dock IR right	2		0 :unreceived 1: received Bit 0 : Receive left emitter data Bit 1 : Receive center emitter data Bit 2 : Receive right emitter data
	Dock IR center	2		0 :unreceived 1: received Bit 0 : Receive left emitter data Bit 1 : Receive center emitter data Bit 2 : Receive right emitter data
	Dock IR left	2		0 :unreceived 1: received Bit 0 : Receive left emitter data Bit 1 : Receive center emitter data Bit 2 : Receive right emitter data

- **Accelerometer RAW Data**

	Name	Size	Value	Description
Header	Identifier	1	0x05	Fixed
Length	Size of data field	1	0x06	Fixed
Data	ACC X axis	2		Unit: 16 bits data, mG
	ACC Y axis	2		Unit: 16 bits data, mG
	ACC Z axis	2		Unit: 16 bits data, mG

- **Gyroscope RAW Data**

	Name	Size	Value	Description
Header	Identifier	1	0x06	Fixed
Length	Size of data field	1	0x06	Fixed
Data	Gyro X axis	2	Int16	Unit: 16 bits data, resolution=2000/32768 (DPS)
	Gyro Y axis	2	Int16	Unit: 16 bits data, resolution=2000/32768 (DPS)
	Gyro Z axis	2	Int16	Unit: 16 bits data, resolution=2000/32768 (DPS)

- **E-compass RAW Data**

	Name	Size	Value	Description
Header	Identifier	1	0x07	Fixed
Length	Size of data field	1	0x06	Fixed
Data	E-compass X axis	2		Unit: 14 bits data, uT
	E-compass Y axis	2		Unit: 14 bits data, uT
	E-compass Z axis	2		Unit: 14 bits data, uT

- **IR Range Finder Information // need to check**

	Name	Size	Value	Description
Header	Identifier	1	0x08	Fixed
Length	Size of data field	1	0x01	Fixed
Data	Drop IR Type	1		Bit 0 : L45 Drop IR Bit 1 : L70 Drop IR Bit 2 : R45 Drop IR Bit 3 : R70 Drop IR Bit 4 : B60 Drop IR 1 for VL53L0 0 for VL6180

- **Motor Temperature Raw Data**

	Name	Size	Value	Description
Header	Identifier	1	0x09	Fixed
Length	Size of data field	1	0x04	Fixed
Data	Temp. Motor #1	1	-128~ 127	Unit: ℃
	Temp. Motor #2	1		Unit: ℃
	Temp. Motor #3	1		Unit: ℃
	Temp. Motor #4	1		Unit: ℃

- **Motor Life**

	Name	Size	Value	Description
Header	Identifier	1	0x0A	Fixed
Length	Size of data field	1	32	Fixed

Data	Life Motor #1	4		Time
		4		Distance
	Life Motor #2	4		Time
		4		Distance
	Life Motor #3	4		Time
		4		Distance
	Life Motor #4	4		Time
		4		Distance

- **Peripheral status for Drop IR**

	Name	Size	Value	Description
Header	Identifier	1	0x0B	Fixed
Length	Size of data field	1	1	Fixed
Data	Status	1		

- **Peripheral status for IMU**

	Name	Size	Value	Description
Header	Identifier	1	0x0C	Fixed
Length	Size of data field	1	1	Fixed
Data	Status	1		

- **Drop IR rate**

	Name	Size	Value	Description
Header	Identifier	1	0x0D	Fixed
Length	Size of data field	1	10	Fixed
Data	Drop IR fifth(back 60)	2	0~65535	Unit: 0.01mcps
	Drop IR fourth(right 70)	2	0~65535	Unit: 0.01mcps
	Drop IR third(right 45)	2	0~65535	Unit: 0.01mcps
	Drop IR second(left 70)	2	0~65535	Unit: 0.01mcps
	Drop IR first (left 45)	2	0~65535	Unit: 0.01mcps

- **Motor current**

	Name	Size	Value	Description
Header	Identifier	1	0x0E	Fixed
Length	Size of data field	1	8	Fixed
Data	Neck yaw current	2	-32768~32767	Unit : mA
Data	Neck pitch current	2	-32768~32767	Unit : mA
Data	Wheel left current	2	-32768~32767	Unit : mA
Data	Wheel right current	2	-32768~32767	Unit : mA

- **Motor Controller Info.**

	Name	Size	Value	Description
Header	Identifier	1	0x0F	Fixed
Length	Size of data field	1	3	Fixed
Type	Info. type	1		2 : motor servo 3 : motor ctrl type
Data		2		**Motor Servo** : 1=servo on, 0=servo off Bit0 : neck yaw Bit1 : neck pitch Bit2 : left wheel Bit3 : right wheel **Motor Ctrl Type**: vel : Bit0~2: 0=navigation, 1=remote, 2=docking, 3=taskremote. Bit3 : 1 pos: Bit0~2: 0=neckremote, 1=smoothpath. Bit3 : 0 which Bit0~3 for neck yaw Bit4~7 for neck pitch Bit8~11 for left wheel

				Bit12~15 for right wheel

- **Hardware Version**

	Name	Size	Value	Description
Header	Identifier	1	0x10	Fixed
Length	Size of data field	1	4	Fixed
Data	Minor	1		0~255
	Major	1		0~255
	ASCII	1		//ASCII 0x52='R'
	ASCII	1	0	//ASCII 0x53='S'

- **Firmware Version**

	Name	Size	Value	Description
Header	Identifier	1	0x11	Fixed
Length	Size of data field	1	4	Fixed
Data	Reserve	1		
	Build	1		
	Minor	1		
	Major	1	0	

- **Hardware_Drop_IR**

	Name	Size	Value	Description
Header	Identifier	1	0x12	Fixed
Length	Size of data field	1	0x01	Fixed
Data	Drop IR Type	1		Bit 0 : L45 Drop IR Bit 1 : L70 Drop IR Bit 2 : R45 Drop IR Bit 3 : R70 Drop IR Bit 4 : B60 Drop IR 1 for VL53L0 0 for VL6180

➢ Controller Limit Info

	Name	Size	Value	Description
Header	Identifier	1	0x13	Fixed
Length	Size of data field	1	0x06	Fixed
Data	Controller type	1		0x00:Velocity Control 0x01:Position Control 0x02:Acceleration
	Controller number	1		0 : neck yaw 1 : neck pitch 2 : left wheel 3 : right wheel
	Upper Limit	2	int16	Max velocity(mm/s)(0.1°/s) for Velocity Control Limit position(0.1°) for Position Control Max Acceleration(mm/s^2) (0.1°/s^2) for Acceleration Value
	Lower Limit	2	int16	Min velocity(mm/s) (0.1°/s) for Velocity Control Limit position(0.1°) for Position Control Min Acceleration(mm/s^2) (0.1°/s^2) for Acceleration Value

➢ Controller Offset Info

	Name	Size	Value	Description
Header	Identifier	1	0x14	Fixed
Length	Size of data field	1	4	Fixed
Data	Controller type	1		0x00:Velocity Control 0x01:Position Control
	Controller number	1		0 : neck yaw 1 : neck pitch 2 : left wheel 3 : right wheel
	Offset	2		

➢ Controller PID Info

	Name	Size	Value	Description
Header	Identifier	1	0x15	Fixed
Length	Size of data field	1	8	Fixed
Data	Controller type	1		0x00:Velocity Control 0x01:Position Control
	Controller number	1		0 : neck yaw 1 : neck pitch 2 : left wheel 3 : right wheel
	P-Gain	2		uint16_t; 1 Unit = 0.01
	I-Gain	2		uint16_t; 1 Unit = 0.1
	D-Gain	2		uint16_t; 1 Unit = 0.0001

➢ Sensor Offset

	Name	Size	Value	Description
Header	Identifier	1	0x16	Fixed
Length	Size of data field	1	2	Fixed
Data	Offset	2		

➢ Sensor EMG Value

	Name	Size	Value	Description
Header	Identifier	1	0x17	Fixed
Length	Size of data field	1	2	Fixed
Data	EMG value	2		

➢ Calibration Values & Status

	Name	Size	Value	Description
Header	Identifier	1	0x18	Fixed
Length	Size of data field	1	0x05	Fixed

	calibration type	1		0x01:Gsensor 0x02: Gyro 0x03:E-compass 0x04:Neck 0x05:Wheel 0x06:Drop_offset 0x07 :Sonar ~~0x08 :Dock(reserve)~~ 0x09:Absolute Encoder 0x0A:Drop_crosstalk 0x0B:Incremental_Encoder 0x0C:Drop 4 line segment (distance) 0x0D:Drop 4 line segment (rate) 0x0E:Position Controller Performance 0x10:Drop IR LEVEL3 distance 0x11:Drop IR LEVEL3 rate 0x0F:Neck Return
	Calibration number	1		0x00: #1 0x01: #2 0x02: #3 0x03: #4 0x04: #5 0x05: #6
Data	Calibration status	1		0x00: fail 0x01: pass
	Calibration data	2		-32767~32766

- **Peripheral Control Feedback**

return low-power control mode

	Name	Size	Value	Description
Header	Identifier	1	0x19	Fixed
Length	Size of data field	1	0x01	Fixed
Data	Mode select	1		Set / Reset. Bit0:sonar process disable Bit1:I2C process disable Bit2:IMU sleep mode Bit3:Drop IR disable

				Bit4:USB update disable Bit5:sonar HW disable

- **Calibration checksum**

	Name	Size	Value	Description
Header	Identifier	1	0x1A	Fixed
Length	Size of data field	1	0x08	Fixed
Data	IMU checksum	1		Checksum of IMU calibration data
	Neck checksum	1		Checksum of Neck calibration data
	Drop_K checksum	1		Checksum of Drop_K calibration data
	Multi segment checksum	1		Checksum of Multi segment calibration data
	reserved	4		reserved

- **Wheel Encoder**

	Name	Size	Value	Description
Header	Identifier	1	0x20	Fixed
Length	Size of data field	1	8	Fixed
Data	Left wheel encoder	4		Accumulated encoder data of left and right wheels in ticks. Increments of this value mean forward direction. It circulates from -2^31 to 2^31 in unit of 0.1degree.
	Right wheel encoder	4		

- **Neck Encoder**

	Name	Size	Value	Description
Header	Identifier	1	0x21	Fixed
Length	Size of data field	1	4	Fixed
	Neck motor yaw Position	2		Accumulated encoder data of 2-axis in 0.1° from -1800 to 1800.
	Neck motor pitch Position	2		

- **Odometry**

(X,Y,theta)

	Name	Size	Value	Description
Header	Identifier	1	0x22	Fixed
Length	Size of data field	1	12	Fixed
Data	**Odometry (x)**	4		The current position of the base in task space in x axis from -2^31 to 2^31 mm
	Odometry (y)	4		The current position of the base in task space in y axis from -2^31 to 2^31 mm
	Odometry (theta)	4		The current heading direction of the base in task space from -2^31 to 2^31 in unit of 0.1degree.

- **Avoid Target**

	Name	Size	Value	Description
Header	Identifier	1	0x23	Fixed
Length	Size of data field	1	8	Fixed
Data	Avoid Target(x)	4		The new target point in x axis while obstacle avoidance(mm) -2^31 to 2^31 mm
	Avoid Target(y)	4		The new target point in y axis while obstacle avoidance(mm) -2^31 to 2^31 mm

- **Neck trajectory**

	Name	Size	Value	Description
Header	Identifier	1	0x30	Fixed
Length	Size of data field	1	4	Fixed
Data	yaw	2		Unit: 0.1deg
	pitch	2		Unit: 0.1deg

- **Wheel trajectory**

	Name	Size	Value	Description
Header	Identifier	1	0x31	Fixed
Length	Size of data field	1	4	Fixed
Data	left	2		Unit: mm/s
	right	2		Unit: mm/s

- **Motor PWM**

	Name	Size	Value	Description
Header	Identifier	1	0x32	Fixed
Length	Size of data field	1	8	Fixed
Data	Yaw	2		Unit: counts
	pitch	2		
	Left	2		
	right	2		

- **Error Code**

	Name	Size	Value	Description
Header	Identifier	1	0xE0	Fixed
Length	Size of data field	1	3	Fixed
Data	Level 0	1	0x00	e_architectureType
	Level 1	1	0x00	
	Level 2	1	0x00	

- **Virtual Data**

	Name	Size	Value	Description
Header	Identifier	1	0xFF	Fixed
Length	Size of data field	1	1	Fixed
Data	Virtual data	1	0x00	